

An exact all-cutoff obstruction to PAH-v2 full-cylinder sup generator convergence

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1. Purpose and scope

This note addresses exactly PAH-V2-GD-001, not a repaired or weaker problem. The immutable preregistration is [strategy/pa-hyp/PAH-v2-GD-001-prereg-v1.json](#), with SHA-256

a3079a674ef328134ed28aafc82d9baa7a16068beb7040410a1a7facca4c3390

The adopted comparison bundle remains

1d6fa47f5351d74f05bed3aba5b1e229a827644d4c6ef5d7dd73c117adee95b6

The separate execution supplement records the user's authorization and fixed probe before the implementation checks. Historical setting-only metadata is preserved. The source PAH-001-v1 functional and PAH-001-v2-r2 labelled state, rates and roots are unchanged; every source hash is in the result certificate.

At fixed parameters and fixed lattice/exhaustion indices h, N , write $I_{r,s}f = f \circ p_{s,r}$. For a single fixed r_0, f_0 in the full finite invariant algebra, the target asks whether

$$\lim_{R \rightarrow \infty} \sup_{s > r \geq R} \|L_s I_{r_0, s} f_0 - I_{r, s} L_r I_{r_0, r} f_0\|_\infty = 0. \quad (1)$$

The norm ranges over every fine counting tuple, not typical Gibbs states. The time is the original external Markov time, without acceleration or a factor of one half in the generator. The original order is local-state cutoff r , lattice h , exhaustion N , then the inherited later stages. Only the first stage is tested; there is no interchange or diagonal limit.

For the counterexample fix $h = N = r_0 = 0$, $q_0 = m_0 = R_0 = 1$, $\epsilon = 1$, and $\beta = \nu = \lambda_s = \eta_6 = \kappa_s = \kappa_D = \kappa_g = g = 1$, $m^2 = \lambda_4 = 0$. These are admitted inserted parameters, not fitted outputs. The literal displayed rates have no additional coefficient. Any separately referenced inserted rate scale is held at unity. The adopted box has side two, four vertices, four stored oriented edges and its single square face, with anchors $O = (0, 0)$ and $C = (1, 0)$. This is an existing combinatorial carrier, not a new physical dimension or volume.

2. Full-source reduction, including all root families

Put $M_r = 2^r$. The remaining prescribed cutoffs are the primorial K_r , $M_{\psi, r} = 4^r$, $R_{\max, r} = 2^r$ and $Q_r = 2^r$. At $\epsilon = 1$,

$$s_v = 1 + j_v(1 - 1)/M_r = 1$$

for every aperture label. Crucially, the approved state space still counts all $j_v \in \{0, \dots, M_r\}$ as distinct. These labels are not quotiented.

The parent functional depends on j only through s . In its onsite terms, $s = 1$ removes every j ; the aperture edge energy is zero; $J_e = 1$ and $J_p = 1$, so the covariant matter and plaquette terms also have no j . An AP move changes no occupation, phase or link coordinate. Therefore every valid AP move has exactly $\Delta F = 0$, mobility $(s_{\text{before}} s_{\text{after}})^{\nu/2} = 1$, and rate

$$c_{\text{AP}} = 1 \exp(-\beta 0/2) = 1. \quad (2)$$

This argument is valid for all other finite coordinates, not only the witness. All finite rates exist: the finite counting space, $\epsilon > 0$ and finite parameters make every displayed functional term and rate finite.

PH changes phase only; LK changes a link only; TR changes occupation only. Consequently they preserve every aperture coordinate and every cutoff pullback of an aperture-only observable. Their increments are separately zero before multiplication by their finite rates. AP at a vertex other than O also has zero increment. These are exact zero contributions in the complete labelled sums, not deletion, matching, averaging or cancellation of roots. Both signs are retained, including the separate duplicate labels at $K = 2$. At $j_O = 0$ the negative AP move is invalid; at $j_O = M_r$ the positive one is invalid. The source omits these moves, without wrapping or reflection.

3. All-index formula and fixed-data counterexample

Fix once and for all $f_0 = \mathbf{1}_{\{j_O=0\}}$ at $r_0 = 0$. It is gauge invariant because gauge transformations leave j fixed, and anchor invariant because the allowed automorphisms fix O . It has full-state sup norm one. R-571 gives, at the same h, N , $j_O \mapsto \lfloor j_O/2^{s-r} \rfloor$. Thus for every $r \geq 0$,

$$f_r = I_{0,r} f_0 = \mathbf{1}_{\{j_O < M_r\}}.$$

This is one fixed cylinder observable, not a cutoff-dependent selection. The complete generator, by (2) and the individual zero increments above, is

$$L_r f_r(x) = \mathbf{1}_{\{j_O=M_r\}} - \mathbf{1}_{\{j_O=M_r-1\}}. \quad (3)$$

Indeed the only downward jump across the indicator discontinuity starts at M_r , and the only upward jump across it starts at $M_r - 1$. The signs in (3) are respectively positive and negative. This also proves the endpoint case $M_r = 1$: the two states have generator values $-1, +1$.

For arbitrary $s > r \geq 0$, set $D = 2^{s-r} \geq 2$, so $M_s = D M_r$. For every fine tuple, the pullback of the positive indicator in (3) is $\mathbf{1}_{\{j_O=M_s\}}$. The pullback of the negative indicator is $\mathbf{1}_{\{M_s-D \leq j_O < M_s\}}$. Subtracting the complete coarse generator from the fine generator gives the exact all-state identity

$$(L_s I_{r,s} f_r - I_{r,s} L_r f_r)(x) = \mathbf{1}_{\{M_s-D \leq j_O < M_s-1\}}. \quad (4)$$

There is no half factor, volume factor, density correction or norm change. The band in (4) is nonempty because $D \geq 2$; it has $D - 1$ integer labels. For any such label a valid full state exists: put all occupation $Q_s = 2^s$ at O , zero occupation elsewhere, and choose any admitted phase/link labels. This is valid since $2^s \leq 4^s$. Hence on the full fine counting space

$$\|L_s I_{r,s} f_r - I_{r,s} L_r f_r\|_\infty = 1 \quad \text{for every } s > r \geq 0. \quad (5)$$

Theorem 1. *The universal full-invariant-cylinder sup-norm Cauchy statement (1) is false for the exact adopted PAH-v2 model and comparison at its allowed $\epsilon = 1$ boundary. Its GD-001 verdict is DISPROVED.*

Proof. Fix the parameters, carrier and observable specified above and $\eta_0 = 1$. For every integer tail threshold $R \geq 0$, choose $r = R$, $s = r + 1$ and $j_O = 2^{r+1} - 2$. Set all other j and every phase/link label to zero and put $\ell_O = Q_s$, all other occupations zero. This is an admitted full tuple at every such cutoff. Formula (4) gives defect one there. Thus there cannot be a tail on which all defects are strictly below η_0 . Formula (5) proves the stronger all-pair norm identity. No finite-table extrapolation enters the argument. $\square$

4. Independent formulation and kernel bridge

An independent row formulation takes the AP generator on $\{0, \dots, M\}$ with off-diagonal entries one between adjacent integers and diagonal equal to minus the number of valid neighbors. Applying this matrix to the vector $(1, \dots, 1, 0)$ gives zero except at its last two coordinates, which are $-1, +1$. Every other full root has zero aperture increment. Pulling back the last two coordinate indicators by integer division gives exactly the two intervals used in (4). This independently audits sign, row sums, endpoint domain and the absence of a multiplicity factor.

The primary executable uses the pinned full source move enumerator and the composed source maps, plus a symbolic expansion of every parent functional term with generic amplitudes and phase cosines. The independent executable imports neither primary nor either source backend: it builds simultaneous coordinate moves, direct all-pair floors and row actions separately. Fixed fixtures check every j_O at $(r, s) = (0, 1), (1, 2), (1, 3), (2, 3)$, including nonzero phase/link labels. These fixtures test the implementations of the general proof; they are not its universal evidence.

[verification/lean/Tect/PahV2GeneratorDefect.lean](#) formalizes five parameterized statements under pinned Lean 4.32.1: endpoint generator, fixed-base pullback, the complete adjacent defect for every $M \geq 1$, its witness at $2M - 2$, and an unbounded-tail witness for every R with $M = 2^R$. The encoded AP definition explicitly omits invalid endpoint moves. The model-to-AP reduction, gauge/anchor invariance and full-state occupation admissibility are the written bridge above, not fully formalized PAH data. The nonadjacent formula (4) is analytic; the Lean adjacent theorem already suffices for the exact negation of (1). No unproved escape is used.

Independent implementation and hostile review were authored within this same task, not by an external referee. External review of (2)–(4), the hidden-label domain and the fixed-data quantifiers is explicitly invited.

5. Devil’s-advocate review and applicability

- **DISMISSED: equal displayed apertures identify the states.** The approved v2 integer counting definition explicitly retains j at $\epsilon = 1$. Quotienting it would change the model and the observable class.
- **DISMISSED: an omitted root cancels the defect.** All PH/TR/LK increments, and AP increments away from O , vanish individually. Both signed AP moves at O are included exactly when valid.
- **DISMISSED: the factor is one half or the sign is reversed.** The source generator has no half; only B has $\sqrt{c/2}$. The definition is fine minus coarse. At the adjacent witness the two values are $0, -1$.
- **DISMISSED: the observable changes with the cutoff.** It is always the same f_0 pulled back from $r_0 = 0$. Only the witness state and the two comparison indices vary, as allowed by the exact negation.
- **DISMISSED: this is only a small-cutoff artifact.** The proof covers every integer $s > r \geq 0$; the kernel covers every adjacent pair through $M = 2^r$, including an explicit witness beyond each threshold.
- **UPHELD boundary: sup failure need not be Gibbs- L^2 failure.** No Gibbs weight or weak convergence was evaluated. This result cannot be used to decide either or a smaller limiting-generator domain.

- **UPHELD boundary: one parameter boundary is not all parameters.** The counterexample disproves the universal target, not every fixed $0 < \epsilon < 1$ case or the finite consistency of R-570/R-571.

The bounded literature/legacy search and hypothesis crosswalk are in the execution supplement. The exact parent definitions supply (2); R-571 supplies the finite floor pullbacks and invariant algebra only. Earlier strip, free-vertex and radial results have different scopes and are not imported. No external theorem or novelty priority is claimed. Units in (3)–(5) are observable per unchanged external Markov time; the zero-tolerance tests are integer/symbolic equalities, not floating approximations.

6. Result footer and one next question

Result ID: R-572; exploration EXP-001739.

Precise statement: Equations (4)–(5) on the allowed $\epsilon = 1$ boundary disprove the exact universal GD-001 target with one fixed f_0 .

Scope: Fixed adopted four-vertex open carrier; arbitrary state cutoff pairs; full fine counting sup norm; original external Markov time.

Dependencies: Immutable PAH-v1 functional, PAH-v2-r2 signed counting definitions, and R-571 adopted comparison; all pins in the certificate.

Evidence grade: ANALYTIC, EXACT, EXECUTED; partial parameterized Lean.

Reproduction command: See the command below and result certificate.

Expected output: All four verification lanes PASS; GD-001 DISPROVED.

Falsification gate: A source-compatible nonzero contribution omitted from (3), an invalid fixed witness, or a failed floor identity would refute this negative result and require append-only correction.

Tier before / after: Host C6 remains T1; no T-054 gate change. The negative result is exact only in its declared model/route scope.

No-overclaim statement: No physical Pre-A, spacetime, QFT, gravity, continuum, common causal cone, Yang–Mills, mass gap or TOE conclusion. No infinite dynamics, alternative-domain no-go or Gibbs- L^2 verdict.

Next required action: Before any replacement convergence target, ask one question: what source-authorized observable domain and topology, if any, should replace the refuted full-cylinder sup generator target while keeping this model and time? No such replacement is selected or tested here.

Reenter this exact target only for a source/domain error or an exact attack on (2)–(4); do not repeat larger finite tables. A proposed weaker target requires separate authorization and must retain this negative record.

```
python -X utf8 verification/scripts/pah_v2_gd001_verify.py
--lean-cache E:/Dev/TECT/verification/lean/.lake/packages
--elan-home C:/Users/NaEun/.elan
```

The displayed command is one line when executed.